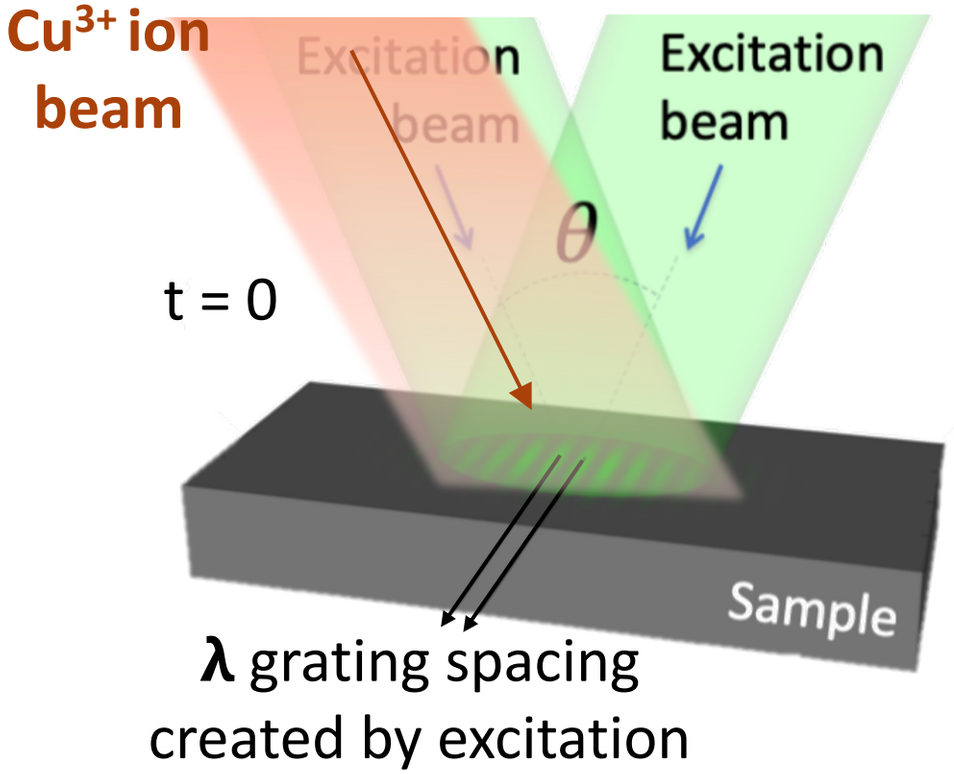
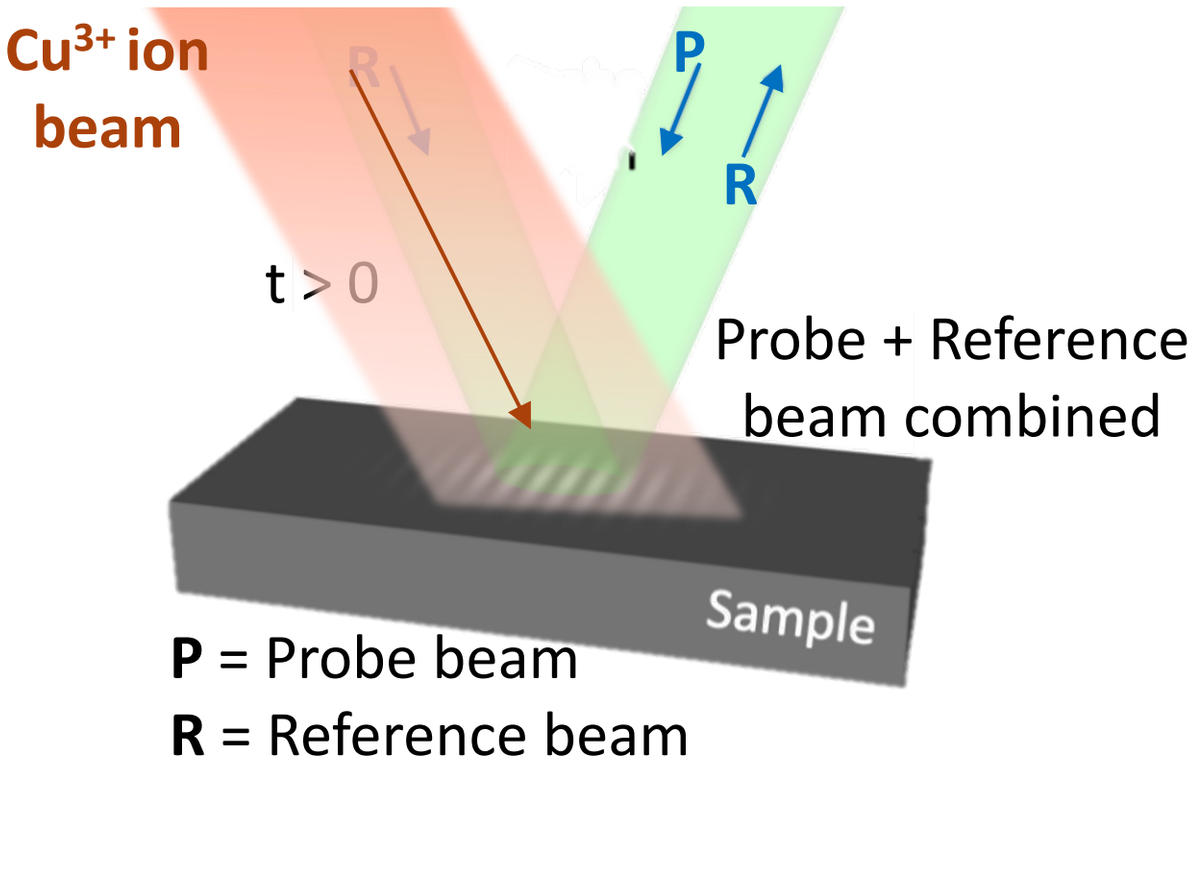
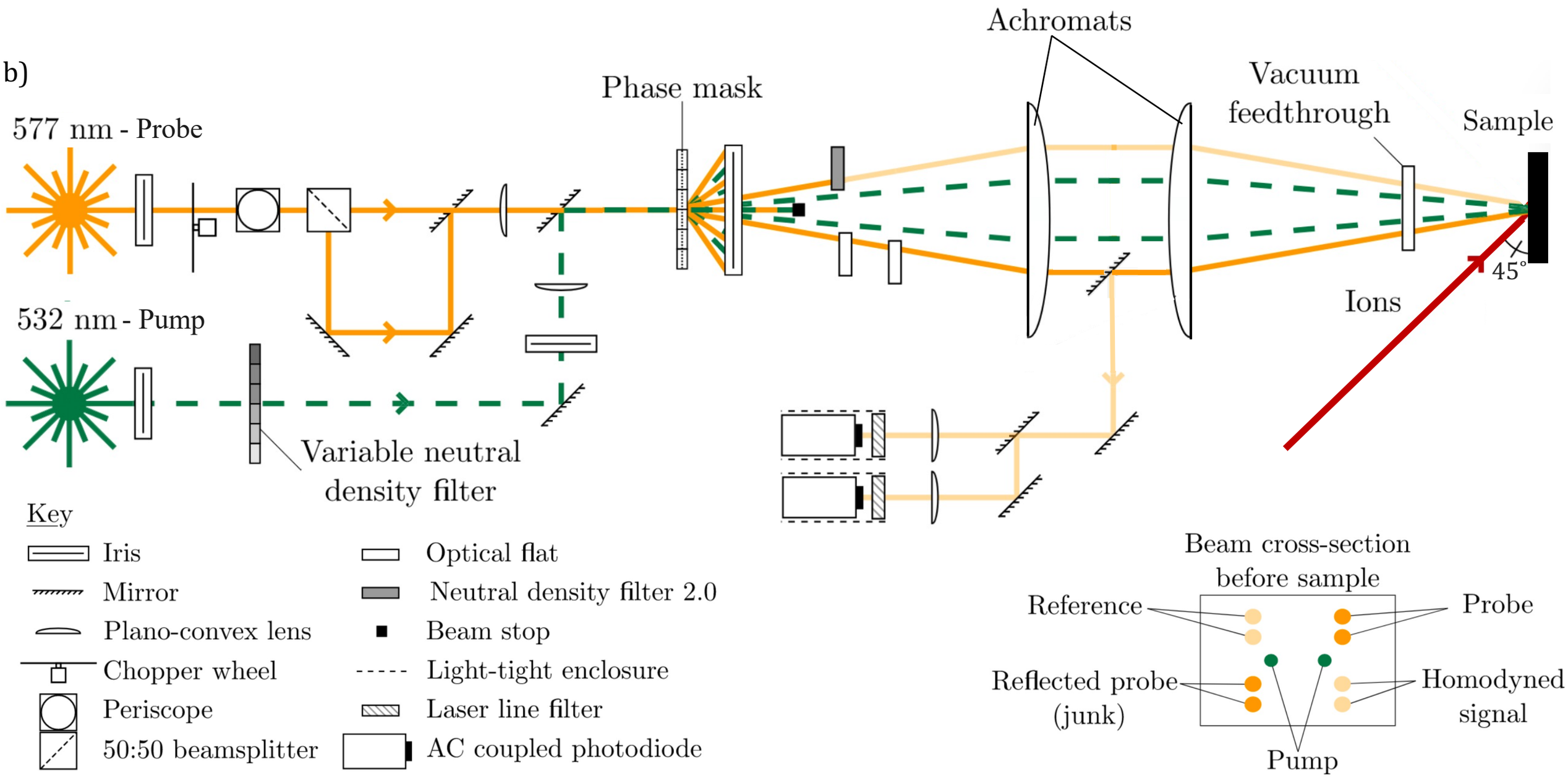
a)


Cu3+ ion beam
Excitation beam
Excitation beam
θ
t = 0
Sample
λ grating spacing created by excitation


Cu3+ ion beam
R
P
R
t > 0
Probe + Reference beam combined
Sample
P = Probe beam
R = Reference beam

b)


Achromats
Phase mask
Vacuum feedthrough
Sample
577 nm - Probe
532 nm - Pump
45°
Ions
Variable neutral density filter
Key
Iris
Mirror
Plano-convex lens
Chopper wheel
Periscope
50:50 beamsplitter
Optical flat
Neutral density filter 2.0
Beam stop
Light-tight enclosure
Laser line filter
AC coupled photodiode
Beam cross-section before sample
Reference
Probe
Reflected probe (junk)
Homodyned signal
Pump